

HEALTHCARE ADMISSIONS

Data Cleaning • Power Query • Star Schema • DAX • Interactive Dashboard

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TOOL	RECORDS	PAGES	ANALYTICAL SCOPE
Microsoft Power BI	54,966	4	Healthcare admissions

PROJECT OVERVIEW

This report documents the development of a four-page healthcare analytics dashboard, covering the documented data-cleaning workflow, derived fields, dimensional model, DAX measures, dashboard structure, design system, and validation checks.

PROJECT OBJECTIVE

Transform cleaned healthcare admission records into an interactive analytical model that supports executive, clinical, financial/insurance, and patient/admission analysis.

DASHBOARD ARCHITECTURE

1	Healthcare Executive Overview — overall admissions, billing, clinical KPIs and trends
2	Clinical Operations Analysis — conditions, test results, age groups and length of stay
3	Finance & Insurance Analysis — billing, insurance providers, financial trends and data quality
4	Patient & Admission Analysis — demographics, admission patterns, medication and outcomes

Document basis: The report is grounded in the supplied project documentation and dashboard screenshots. Unsupported assumptions are intentionally excluded.

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1. Dataset Overview

The project uses a public synthetic healthcare dataset from Kaggle. It is randomly generated for practice/portfolio use and is not real patient data.

Item	Documented value
Source	Kaggle synthetic "Healthcare Dataset"
Original size	55,500 rows × 15 columns
Final cleaned size	54,966 rows
Exact duplicates removed	534
Date range	2019-05-08 to 2024-05-07
Billing range in source	-\$2,008 to \$52,764

Original columns

Column	Type / treatment	Description
Name	Text	Patient name
Age	Whole Number	Patient age; documented range 13–89
Gender	Text	Male / Female
Blood Type	Text	8 blood types
Medical Condition	Text	6 conditions
Date of Admission	Text → Date	Admission date
Doctor	Text	Attending doctor
Hospital	Text	Hospital/facility
Insurance Provider	Text	5 providers
Billing Amount	Decimal	Billed amount
Room Number	Whole Number	Room assigned
Admission Type	Text	Urgent / Emergency / Elective
Discharge Date	Text → Date	Discharge date
Medication	Text	5 medications
Test Results	Text	Normal / Inconclusive / Abnormal

2. Data Quality Assessment

Issue	Observed problem	Decision / treatment
Duplicate rows	534 exact duplicate rows	Removed
Inconsistent name casing	Random capitalization such as "BobbY JacksOn"	Standardized to proper case
Negative billing	Minimum -\$2,008; no transaction/reason field	Billing_Flag = Invalid; excluded from billing totals
Billing precision	Up to 15 decimal places	Rounded to 2 decimals
Dates stored as text	Admission/discharge dates were text	Converted to Date
No unique identifier	No original patient/record ID	PatientRecordID added after deduplication
Hospital text noise	Leading "and " / trailing commas; some word-order scrambling	Leading/trailing noise partially cleaned; internal scrambling retained
Doctor/Hospital cardinality	~40,000 distinct values each	Kept in fact table; no separate dimensions

[Important] Negative billing is treated as data-quality noise rather than a legitimate refund/adjustment because the dataset has no supporting transaction-type or reason-code field.

3. Power Query Data Cleaning Process

1. Promoted headers and changed column types.
2. Removed duplicates using all columns, before adding the index.
3. Capitalized each word in Name and Doctor.
4. Trimmed and cleaned text columns.
5. Applied custom cleaning to Hospital.
6. Created Billing_Flag with Valid / Invalid based on Billing Amount sign.
7. Converted Date of Admission and Discharge Date to proper Date type.
8. Created LengthOfStay from discharge date minus admission date.
9. Created AgeGroup using the documented age bands.
10. Added PatientRecordID as an Index from 1, after deduplication.
11. Confirmed final cleaned row count of 54,966.

Hospital cleaning expression

```
Text.Trim(Text.TrimEnd(if Text.StartsWith([Hospital], "and ") then Text.RemoveRange([Hospital], 0, 4) else [Hospital], {"", "}))
```

4. Created / Derived Columns

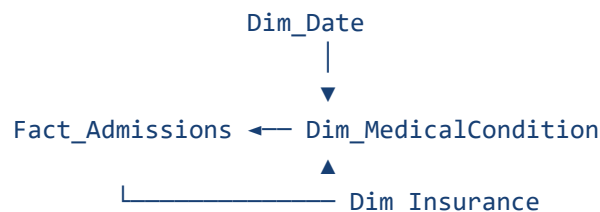
Column	Created in	Logic / purpose
Billing_Flag	Power Query	Valid for non-negative Billing Amount; Invalid for negative values
LengthOfStay	Power Query	Duration.Days(Discharge Date – Date of Admission)
AgeGroup	Power Query	Minor; Adult (18-39); Middle-Aged (40-59); Senior (60+)
PatientRecordID	Power Query	Index from 1, added after deduplication
MonthStart	DAX	First day of month; used for chronological sorting
MonthYear	DAX	FORMAT(MonthStart, "MMM YYYY"); sorted by MonthStart

AgeGroup logic

```
if [Age] < 18 then "Minor"
else if [Age] < 40 then "Adult (18-39)"
else if [Age] < 60 then "Middle-Aged (40-59)"
else "Senior (60+)"
```

5. Star Schema and Data Model

The documented model uses Fact_Admissions as the central fact table with Dim_MedicalCondition, Dim_Insurance and Dim_Date.



Fact_Admissions

PatientRecordID | Name | Age | Gender | Blood Type | ConditionID | Date of Admission | Doctor | Hospital | InsuranceID | Billing Amount | Billing_Flag | Room Number | Admission Type | Discharge Date | LengthOfStay | Medication | Test Results | AgeGroup

Dimensions

Table	Rows	Construction / columns
Dim_MedicalCondition	6	ConditionID, Medical Condition; Reference → Remove Other Columns → Remove Duplicates → Index
Dim_Insurance	5	InsuranceID, Insurance Provider; same construction pattern
Dim_Date	Date range	DAX calendar table; marked as official Date Table

Relationships

Fact field	Dimension field	Relationship
ConditionID	Dim_MedicalCondition[ConditionID]	Many-to-One, Single direction
InsuranceID	Dim_Insurance[InsuranceID]	Many-to-One, Single direction
Date of Admission	Dim_Date[Date]	Many-to-One, Single direction

The raw healthcare_dataset_original table is intentionally disconnected and retained for audit/comparison.

[Modeling choice] Doctor and Hospital were not modeled as dimensions because their documented cardinality is approximately 40,000 distinct values each across the cleaned data.

6. Date Table and Time Intelligence

```
Dim_Date =
ADDCOLUMNS(
    CALENDAR(MIN(Fact_Admissions[Date of Admission]), MAX(Fact_Admissions[Discharge Date])),
    "Year", YEAR([Date]),
    "MonthNumber", MONTH([Date]),
    "MonthName", FORMAT([Date], "MMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "YearMonth", FORMAT([Date], "MMM YYYY"),
    "DayOfWeek", FORMAT([Date], "ddd")
)
```

Dim_Date is marked as the official Date Table using Date.

Chronological month sorting

```
MonthStart = DATE(YEAR(Dim_Date[Date]), MONTH(Dim_Date[Date]), 1)
MonthYear = FORMAT(Dim_Date[MonthStart], "MMM YYYY")
```

MonthYear is sorted by MonthStart to prevent alphabetical month ordering.

7. DAX Measures

7.1 Core

```
Total Admissions = COUNTROWS(Fact_Admissions)

Total Billing =
CALCULATE(
    SUM(Fact_Admissions[Billing Amount]),
    Fact_Admissions[Billing_Flag] = "Valid"
)

Avg Length of Stay = AVERAGE(Fact_Admissions[LengthOfStay])

Avg Billing per Admission =
DIVIDE(
    [Total Billing],
    CALCULATE(COUNTROWS(Fact_Admissions), Fact_Admissions[Billing_Flag] = "Valid")
)

Avg Age = AVERAGE(Fact_Admissions[Age])
```

7.2 Rates / data quality

```
Abnormal Test Rate % =
DIVIDE(
    CALCULATE([Total Admissions], Fact_Admissions[Test Results] = "Abnormal"),
    [Total Admissions]
)

Emergency Admission Rate % =
DIVIDE(
    CALCULATE([Total Admissions], Fact_Admissions[Admission Type] = "Emergency"),
    [Total Admissions]
)

Billing Data Quality % = -- documented as invalid-record rate
DIVIDE(
    CALCULATE(COUNTROWS(Fact_Admissions), Fact_Admissions[Billing_Flag] = "Invalid"),
    COUNTROWS(Fact_Admissions),
    0
)
```

```

Valid Billing % = -- documented as valid-record rate
DIVIDE(
    CALCULATE(COUNTROWS(Fact_Admissions), Fact_Admissions[Billing_Flag] = "Valid"),
    COUNTROWS(Fact_Admissions),
    0
)

Invalid Billing Records =
CALCULATE(
    COUNTROWS(Fact_Admissions),
    Fact_Admissions[Billing_Flag] = "Invalid"
)

```

[Naming caution] Billing Data Quality % is mathematically the invalid-record rate in the supplied formula. If the intended KPI is the percentage of valid records, the measure name should be changed rather than silently changing the documented formula.

7.3 Time intelligence

```

YoY Admissions Growth % =
VAR CurrentPeriod = [Total Admissions]
VAR PriorPeriod = CALCULATE([Total Admissions], SAMEPERIODLASTYEAR(Dim_Date[Date]))
RETURN DIVIDE(CurrentPeriod - PriorPeriod, PriorPeriod)

YoY Billing Growth % =
VAR CurrentPeriod = [Total Billing]
VAR PriorPeriod = CALCULATE([Total Billing], SAMEPERIODLASTYEAR(Dim_Date[Date]))
RETURN DIVIDE(CurrentPeriod - PriorPeriod, PriorPeriod)

Running Total Billing =
CALCULATE(
    [Total Billing],
    FILTER(ALLSELECTED(Dim_Date[Date]), Dim_Date[Date] <= MAX(Dim_Date[Date]))
)

```

7.4 Ranking

```

Top Condition by Cost Rank =
RANKX(
    ALL(Dim_MedicalCondition[Medical Condition]),
    CALCULATE([Total Billing]),
    ,
    DESC
)

```

8. Documented Dashboard Structure

The supplied textual project documentation describes three dashboard pages. Later screenshots show a fourth page, but its final specification is not present in the supplied text; therefore no unsupported Page 4 definition is asserted here.

Page	Purpose	Documented visuals
Page 1 — Executive Overview	Management overview	KPI cards: Total Admissions, Total Billing, Avg Length of Stay, Abnormal Test Rate %; Total Billing line chart by YearMonth; Admission Type donut; Medical Condition × Total Billing table with data bars
Page 2 — Clinical Analysis	Clinical patterns	KPI cards: Total Admissions, Avg Length of Stay, Abnormal Test Rate %; Total Billing by Medical Condition bar chart; Medical Condition × Test Results matrix; Decomposition Tree: Condition → Admission Type → Gender
Page 3 — Financial / Insurance Analysis	Financial and insurance analysis	KPI cards: Total Billing, Avg Billing per Admission, Valid Billing %, Invalid Billing Records; Age vs Billing Amount scatter/bubble by Medical Condition; Avg Billing per Admission by Insurance Provider; optional Insurance Provider rank-over-time ribbon

9. Design System

Element	Documented specification
Canvas background	#FFFFFF / #FAFAFA
Header bar	#1B4D3E with white text
Primary data color	#3E7C59
Secondary data color	#E0A458
Card background	White; 1px #E5E7EB border; no shadow; 8px radius
Primary text	#1F2937
Secondary labels	#6B7280
KPI value	28–32 pt, bold, primary accent
Chart titles	Plain-English insight questions; 13 pt semibold; left aligned
Spacing	12–16 px between visuals; 20 px page margins

10. Validation Checklist

- Duplicate removal happened before PatientRecordID creation.
- Final cleaned row count is 54,966.
- Admission and discharge dates are Date types.
- LengthOfStay is based on discharge minus admission date.
- AgeGroup follows the documented four age bands.
- Billing_Flag separates negative billing records from valid billing records.
- Total Billing filters Billing_Flag = Valid.
- Avg Billing per Admission uses valid billing records in its denominator.
- Dim_Date is marked as the official Date Table.
- MonthYear is sorted by MonthStart.
- All three documented relationships are Many-to-One and single-direction.
- healthcare_dataset_original remains disconnected.
- Doctor and Hospital remain detail attributes in Fact_Admissions.

11. Known Limitations and Cautions

- Hospital and Doctor names contain synthetic-data word-order scrambling. Only leading 'and ' and trailing commas were corrected.
- Negative billing values are treated as data-quality noise and excluded from billing totals because no transaction-type/reason field supports a refund interpretation.
- Doctor and Hospital are not dimensions because of near-unique cardinality and limited aggregation benefit.
- The textual source does not fully specify the later four-page dashboard shown in screenshots, so this document does not invent the missing Page 4 definition.
- The measure named Billing Data Quality % is documented as an invalid-record rate; review the name if the intended meaning is valid-record percentage.

Appendix — Project Outcome Summary

Stage	Result
Raw dataset	55,500 rows × 15 columns
Deduplication	534 exact duplicates removed
Cleaned dataset	54,966 rows
Power Query additions	Billing_Flag, LengthOfStay, AgeGroup, PatientRecordID
Dimensions	Dim_Date, Dim_MedicalCondition, Dim_Insurance
Fact table	Fact_Admissions
Date intelligence	Year, MonthNumber, MonthName, Quarter, YearMonth, DayOfWeek, MonthStart, MonthYear
DAX	Core, quality/rates, time intelligence, ranking

End of documented process

12. Final Dashboard Visualization Review

The following four screenshots document the final dashboard pages supplied with the project. This section records what each page is designed to communicate, the role of its visuals, and the main analytical questions a viewer can answer. The descriptions are based on the visible dashboard content; they do not introduce unsupported business conclusions.

PROJECT OVERVIEW

Project: Healthcare Admissions & Financial Analytics Dashboard | Tool: Microsoft Power BI | Total Records: 54,966 | Dashboard Pages: 4

Objective: Analyze healthcare admissions, patient demographics, clinical outcomes, billing performance, insurance-provider performance, and admission patterns through an interactive Power BI dashboard.

Dashboard structure: 1) Healthcare Executive Overview • 2) Clinical Operations Analysis • 3) Finance & Insurance Analysis • 4) Patient & Admission Analysis

Key analytical areas: Admissions • Billing • Insurance • Patient Demographics • Medical Conditions • Test Results • Length of Stay • Medication

12.1 Page 1 — Healthcare Executive Overview

Figure 12.1 — Final Executive Overview dashboard

Purpose: provide a management-level snapshot of admissions, billing, length of stay, test-result quality, and admission patterns, with slicers for Year, Hospitals, Admission Type, Insurance Provider, and Medical Condition.

What this page communicates:

- Four headline KPIs: Total Admissions (55K), Total Billing (\$1.4bn), Avg Length of Stay (15.50), and Abnormal Test Rate (33.54%).
- The Total Billing Trend over Time combines monthly billing bars with a YoY Billing Growth % line, allowing trend and growth to be viewed together.
- Admissions by Type shows a near-even split among Elective (32.93%), Urgent (33.46%), and Emergency (33.61%) admissions.
- The condition table combines Total Admissions, Total Billing, Avg Billing per Admission, Avg Length of Stay, and Abnormal Test Rate % in one comparison view.

Analytical questions it supports:

- How large is the admission and billing volume?
- How is billing changing over time?
- How evenly are admissions distributed across admission types?
- Which medical conditions have higher or lower billing, length of stay, or abnormal-test rates?

12.2 Page 2 — Clinical Operations Analysis

Figure 12.2 — Final Clinical Operations Analysis dashboard

Purpose: focus on patient volume, length of stay, test-result distribution, age groups, admission type, and condition-level clinical metrics.

What this page communicates:

- The KPI row reports Total Admissions (55K), Avg Length of Stay (15.50), Abnormal Test Rate (33.54%), and Emergency Admission Rate (32.93%).
- Average Length of Stay by Medical Condition compares the six medical conditions using a horizontal bar chart.
- Test Result Distribution by Medical Condition breaks each condition into Abnormal, Inconclusive, and Normal results using a 100% stacked bar chart.
- Admissions by Age Group compares Adult (18–39), Middle-Aged (40–59), Minor, and Senior (60+) groups.
- The lower visuals provide a second view of admission type and a detailed condition × test-result count matrix.

- The Key Metrics by Medical Condition table combines admissions, average length of stay, and abnormal test rate for direct condition-level comparison.

Analytical questions it supports:

- Which conditions have relatively higher average length of stay?
- How are test results distributed within each medical condition?
- Which age groups account for the largest admission volumes?
- How do clinical metrics differ across medical conditions?

12.3 Page 3 — Finance & Insurance Analysis

Figure 12.3 — Final Finance & Insurance Analysis dashboard

Purpose: analyze billing performance, insurance-provider differences, billing validity, cost relationships, and billing trends over time.

What this page communicates:

- Headline KPIs show Total Billing (\$1.4bn), Avg Billing per Admission (\$25.6K), and Valid Billing Rate (99.81%).
- Billing Data Quality separates valid and invalid billing records; the visible invalid share is 0.19%.
- Total Billing Trend & YoY Growth combines annual billing with year-over-year growth.
- Average Billing per Admission by Insurance Provider compares the five insurance providers on a common cost-per-admission basis.
- Billing Cost vs Length of Stay by Medical Condition examines the relationship between two average metrics at condition level, with bubbles representing medical conditions.
- Total Billing by Insurance Provider compares overall billing contribution across providers.
- Insurance Provider Billing Trend shows provider-level billing movement over time.
- The Financial Performance by Medical Condition table combines admissions, billing, average billing per admission, length of stay, share of total billing, and abnormal-test rate.

Analytical questions it supports:

- What is the overall billing volume and average billing per admission?
- Which insurance providers have higher average billing per admission?
- How much of the billing data is currently classified as valid?
- How does billing differ among medical conditions?
- How do insurance-provider billing patterns change over time?

12.4 Page 4 — Patient & Admission Analysis

Figure 12.4 — Final Patient & Admission Analysis dashboard

Purpose: provide a patient-demographic and admission-pattern view using age, gender, blood type, medication, medical condition, admission type, and test-result information.

What this page communicates:

- Headline KPIs show Total Admissions (55K), Avg Age (51.54), Female Patient % (49.98%), and Unique Hospital Names (40K).
- Admissions by Age Group & Admission Type compares age groups while splitting each group by admission type.
- Patient Distribution by Gender shows the male/female distribution; the visible values are approximately 50% each.
- Patient Distribution by Blood Type compares all eight blood-type categories.
- Patient Distribution by Medication compares the five medication categories by admission count.
- Admissions by Medical Condition uses a treemap to show the relative admission volume of the six medical conditions.
- Test Results by Admission Type compares Normal, Inconclusive, and Abnormal proportions for Elective, Urgent, and Emergency admissions.
- The Patient Profile by Medical Condition table combines admissions, average age, average length of stay, average billing per admission, and abnormal-test rate.

Analytical questions it supports:

- What is the overall demographic profile of the admissions?
- How are admissions distributed by age group, gender, blood type, and medication?
- How do admission types vary across age groups?
- How are medical conditions distributed by admission volume?
- Do test-result proportions differ across admission types?
- How do patient and admission metrics vary by medical condition?

12.5 Cross-Page Analytical Story

The four pages form a layered analytical flow rather than four unrelated dashboards:

- Executive Overview → establishes the overall scale and major KPIs.
- Clinical Operations → explains admission and patient-condition patterns from an operational/clinical perspective.
- Finance & Insurance → focuses on billing quality, cost, insurance-provider performance, and financial trends.
- Patient & Admission Analysis → provides demographic and admission-detail exploration.

12.6 Visual-to-Question Mapping

Visual type	Used for	Primary analytical purpose
KPI cards	Executive / clinical / financial headline metrics	Quickly establish scale and current performance
Line / combo chart	Billing over time and YoY growth	Trend and growth analysis
Donut chart	Admission type, gender, billing validity	Part-to-whole comparison
Bar chart	Length of stay, insurance billing, blood type, medication	Ranked category comparison
100% stacked bar	Test results by condition or admission type	Compare proportions rather than raw totals
Treemap	Admissions by medical condition	Compact part-to-whole comparison across conditions
Scatter / bubble chart	Billing vs length of stay by condition	Examine relationship between two measures
Tables / matrices	Condition-level clinical and financial metrics	Detailed cross-category comparison
Area / flow-style trend	Insurance provider billing over time	Compare provider contribution across periods

12.7 Interpretation Boundaries

- The dashboards are descriptive/diagnostic visualizations of the supplied synthetic dataset. They should not be interpreted as evidence about real hospitals or real patients.
- The visuals show associations and distributions; they do not by themselves establish causation.
- The dashboard uses the documented Billing_Flag rule, so financial totals are based on records classified as valid billing.
- The final screenshots contain a fourth page, Patient & Admission Analysis, which is now documented here from the supplied screenshot even though the earlier textual project notes did not fully specify that page.
- Values displayed in KPI cards are rounded for presentation (for example 55K and \$1.4bn); detailed tables contain more precise values.